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# New paleomagnetic results from the Eureka Sound Group: Implications for the age of early Tertiary Arctic biota

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## ABSTRACT

**Some 300 specimens from 101 sites were analyzed from sections spanning more than 2,000 m of lignitic carbonate and clastic rocks. The sediments belong to the lower Tertiary Eureka Sound Group and are exposed near Bay and Strathcona Fiords on Ellesmere Island. Paleomagnetic analysis suggests that remanence is carried by magnetite, and positive fold and reversals tests support an early (detrital or early post-depositional) acquisition of the characteristic magnetization. We correlate the local magnetostratigraphy to Chrons C25 and C26 of the time scale. Comparison of the magnetostratigraphic results obtained here with results from Paleocene/Eocene sections from Wyoming and Texas do not support a significant temporal discrepancy of faunal and floral assemblages between the Arctic and the southwestern United States.**

## INTRODUCTION

Vertebrate-fossil–collecting expeditions to the terrestrial sediments of the Eureka Sound Group, exposed on several Canadian islands north of the Arctic Circle, were originally inspired as a search for a land bridge connecting the North American and European mammalian faunas (West and Dawson, 1978). Such a bridge was thought to exist during the early and middle Eocene, owing to a striking similarity of Wasatchian North American and Sparnacian European mammalian faunas. The connection was presumably severed later in the Eocene by continued sea-floor spreading which resulted in a biographic separation of the two faunas (see McKenna, 1975).

The results of these exploratory expeditions stirred controversy on several fronts. The discovery of vertebrates such as crocodiles and turtles from outcrops within hundreds of miles of the North Pole, which in modern communities suggests warm and wet environments, led to a variety of possible explanations. These included changing the spin axis of the Earth, the solar constant, or the Earth's obliquity to name a few (see McKenna, 1980, for review). Furthermore, Hickey and others (1983) reported an apparent age disparity of the entire Wasatchian mammalian fauna of about 2 m.y. and of the palynological data of as much as 18 m.y. between the Arctic and North American records. The assertion by Hickey and others (1983) that the Arctic fauna and flora predated similar populations found elsewhere in North America was based primarily on the unpublished paleomagnetic

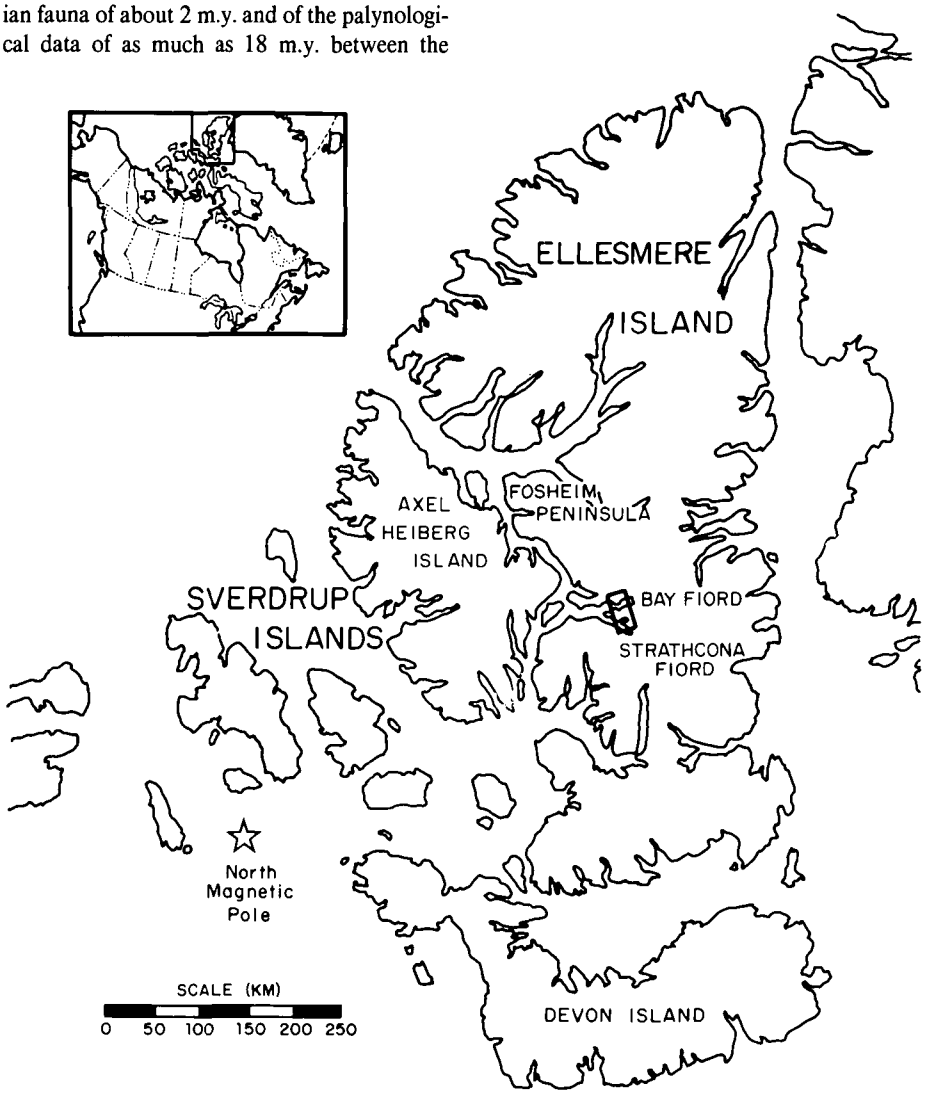



**Figure 1. Map of northern Canadian islands showing location of Ellesmere and Axel Heiberg Islands. Small box on Ellesmere shows location of field area and is enlarged in Figure 2.**

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data of Vinson (1981), the reliability of which was questioned by Kent and others (1984). Flynn and others (1984) analyzed the theoretical principles and concluded that such heterochrony is unlikely. Furthermore, their analysis of postulated cases suggested that there is no strong evidence supporting faunal heterochrony.

In order to resolve the controversy over the paleomagnetic data quoted in Hickey and others (1983), we participated in an expedition to Ellesmere Island in July 1985. We present here the results of our paleomagnetic investigations in an attempt to clarify the temporal relationship of the Ellesmere outcrops to those found in the southwestern United States (summarized by Rapp and others, 1983; Flynn and others, 1984; Butler and Lindsay, 1985).

## GEOLOGICAL BACKGROUND

Collections of the flora and fauna of the Canadian Arctic come from the molassic terrestrial and marine deposits described as informal members of the Eureka Sound Formation (Tozer, 1963; Miall, 1981; West and others, 1981). Miall (1986) redefined these sediments as the Mount Lawson, Mount Moore, and Margaret Formations of the Eureka Sound Group. An alternative definition was recently published by Ricketts (1986). We use the terminology of Miall (1986) because of precedence in the literature.

The lignitic clastics and carbonates of the Eureka Sound Group have been mapped in seven depositional basins over a broad area but are principally found on Axel Heiberg and Ellesmere Islands (Fig. 1). Mammalian collections come exclusively from outcrops on Ellesmere Island near Bay and Strathcona Fiords (see Figs. 1 and 2; Dawson and others, 1976; West and others, 1977, 1981; West and Dawson, 1978). These outcrops were the focus of our preliminary paleomagnetic investigations.

The sediments of the Bay Fiord area, shown in Figure 2, were divided into four informal members by West and others (1981) and subsequently elevated to formational status by Miall (1986). These formations have been described as lithofacies assemblages and interpreted in terms of depositional environment by Miall (1981, 1986) and West and others (1981). We will briefly describe the geology here.

The base of the Eureka Sound Group conformably overlies the Kanguk Formation in the Bay Fiord area. The lowest formation, the Mount Bell Formation (member I of West and others, 1981) comprises lenses of light, poorly consolidated, arkosic to quartzose, cross-bedded sandstone within a flaggy sandstone and gray mudstone matrix. These deposits were interpreted by West and others (1981) as bottomset beds and lower delta plain facies.

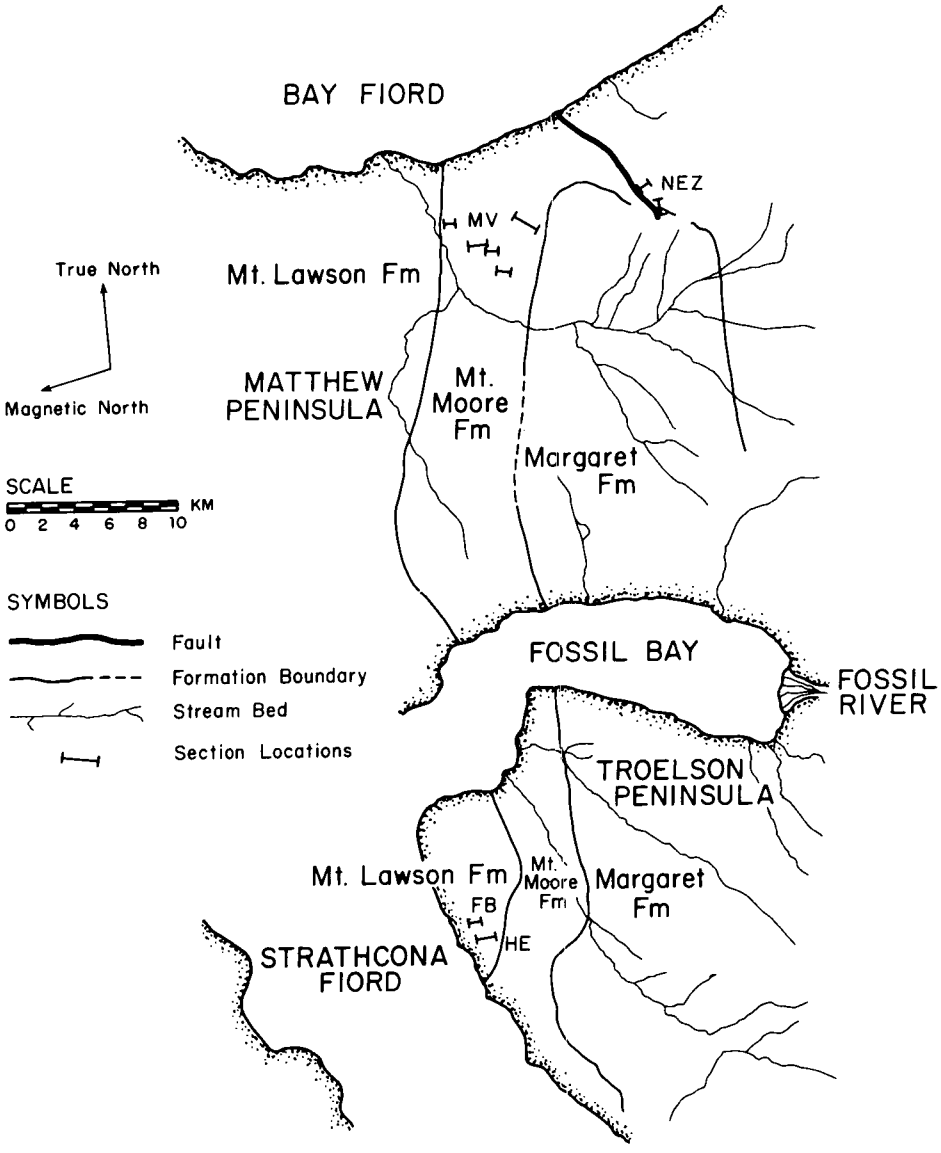



**Figure 2. Map showing location of sections on Matthew and Troelson Peninsula.**

The Mount Bell Formation (member I), 105 m thick in its type area, gives way to the widespread Mount Lawson Formation (member II), quoted as being as much as 1,500 m thick and of Late Cretaceous to Paleogene age (West and others, 1981). The lower part of this formation is dominated by light, poorly consolidated lenses of medium to coarse sandstone in a matrix of dark gray mudstone, shale, and coal. The upper third is composed of well-bedded, dark gray siltstones and claystones, punctuated by occasional thin sandstone and conglomerate units. West and others (1981) suggested a fluvial origin for the lower portion and a swamp origin for the upper third. Miall (1981, 1986) described it as a distal delta-front facies. The Mount Lawson Formation is conformably overlain by the Mount Moore Formation (formerly known as member III), a cream-colored unit composed of marl, limestone, and sandstone. Miall (1981, 1986) reported estuarine to shallow-marine deposition, whereas West and others (1981) suggested a lagoonal origin for the lower part, fully marine conditions for the middle, and shallow marine to brackish-water deposition for the top. West and others (1975) described Paleocene to Eocene marine fossils from deposits of member III aspect. The temporal relationship between those deposits and ours is not clear, however.

The mammalian fossils are all found in the Margaret Formation (member IV), which is

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characterized by thick (up to 15 m) coal beds interbedded with unconsolidated, well-sorted, light gray to white sandstones and infrequent thin shale. The environment of deposition of these sediments has been interpreted as proximal delta front and delta plain (Miall, 1981, 1986). The mammals come from two levels, the lower of which has been classified as late Wasatchian North American Land Mammal Age and the upper level could be as young as early Bridgerian (see McKenna, 1980).

TABLE 1. SECTION INFORMATION

| Section | Location | Formation | No. of sites | Thickness (m) |
|---|---|---|---|---|
| FB, HE | 78°41.2'N, 82°32'W | Mount Lawson | 7, 33 | 490 |
| MV | 78°49.4'N, 82°27'W | Mount Moore | 50 | 1,332 |
| NEZ | 78°49.5'N, 82°17.5'W | Mount Moore/Margaret | 11 | 216 |

## FIELD METHODS

All paleomagnetic specimens were taken as hand samples owing to the logistical difficulties associated with gasoline-powered, water-cooled "portable" drills. Shallow pits were dug with a pick-mattock in order to remove the debris resulting from frost heaving. Permafrost was generally found from 1–2 ft below the surface, limiting our access to fresh material. Whenever possible, samples were taken from recent cuts made by spring runoff, as these yielded the least-disturbed material. A flat face was shaved on the sample with a hand rasp. A horizontal arrow and the direction of dip were marked on the sample. The direction of the arrow with respect to magnetic North and the angle of the dip were measured using an International Brunton Compass. A minimum of three separately oriented samples were taken at each site in order to establish within-site consistency.

The field area is less than 500 km away from the magnetic north pole (Fig. 1). The reliability of the International Brunton was established in three ways. First, daily measurements were made at the base camp along a line oriented by air photos. These varied by about ±4°, presumbly the result of magnetic storms. The base-camp measurements suggested an average magnetic declination of 257°. Second, five sun compass readings were taken using a protractor and a plumb-bob along with Brunton measurements on a similarly oriented line. These were: 12°, 18°, 8°, 3°, and 3° with a mean of 9°. The variability was caused by the effect of wind on the make-shift plumb-bob. The mean compass measurement was 102°, suggesting a magnetic declination for the day of 267° with a combined error of about ±10°. Third, each sample was separately oriented. Care was taken to remove all magnetic objects from the area. Consistency of paleomagnetic directions among samples from the same site is a requirement of our data-selection procedure and lends credence to the compass measurements. Furthermore, care was taken to check measurements against clues given by landmarks on air photographs. We estimate the error in direction of strike of the orientation face to be ±10° and the error in the dip angle of the face to be ±1°. A local strike and dip of the beds was measured for each site. A declination correction of 103° was subtracted from all declination measurements prior to analysis; all data presented here are thus referenced to true North. Stratigraphic thicknesses were measured using a Jacob's staff and an Abney level. The location of each site was marked on an air photograph and documented using a Polaroid instant camera for ease of future relocation and integration with prior work.

A total of four sections were taken in the Mount Lawson, Mount Moore, and the lowermost Margaret Formation (see Table 1 and Fig. 2). Although Hickey and others (1983) reported results from the Mount Bell Formation (Member I), we could not locate any outcrops which we deemed suitable for paleomagnetic sampling during our preliminary field season. Our section

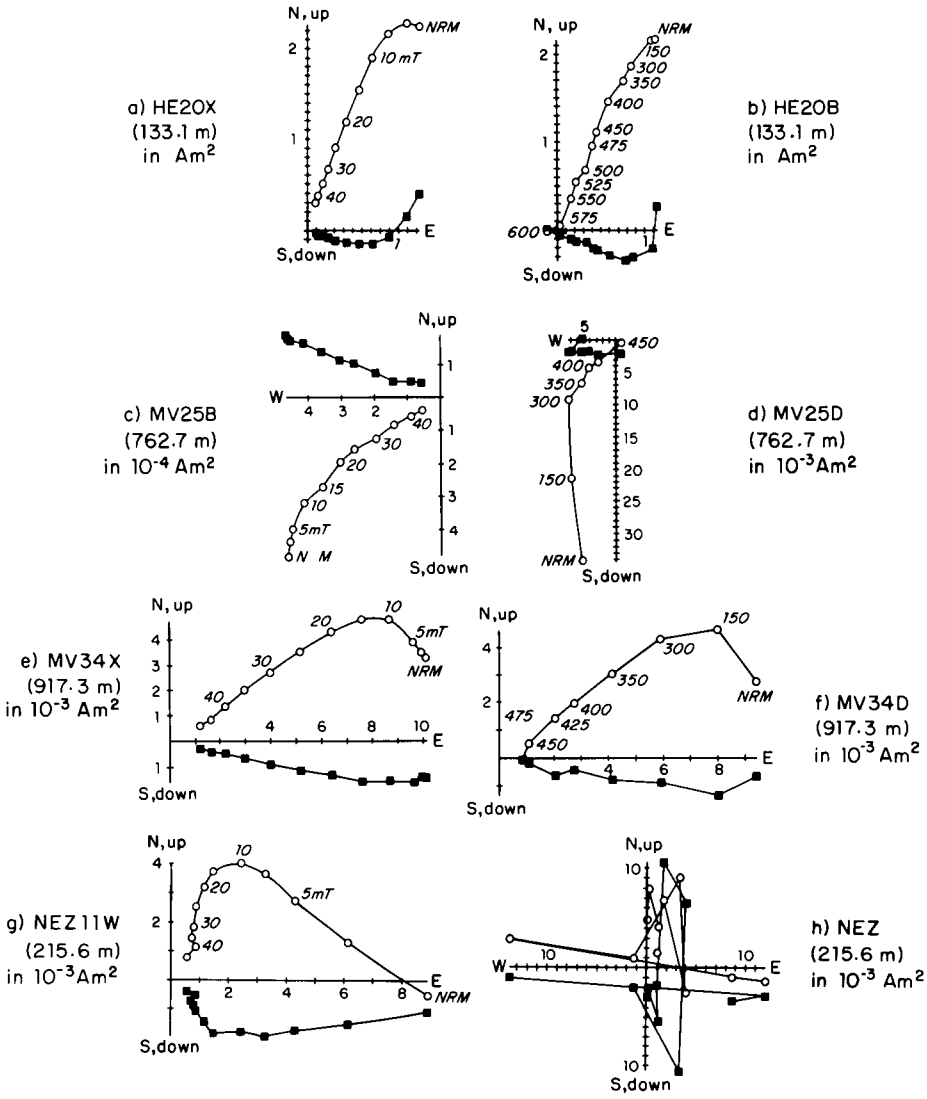



**Figure 3. Vector end-point diagrams. Solid squares are projections onto the horizontal plane; open circles are projections onto the vertical, east-west plane.**

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locations, formation, number of sites, and total stratigraphic thickness are listed in Table 1.

Two partially overlapping sections were taken in the Mount Lawson Formation, HE and FB; these were separated laterally by ~100 m of continuous exposure. Sampling was terminated at the base by the lower, sandy facies of this formation (considered unsuitable for paleomagnetic analysis) and at the top by poor exposure. The top of the section is estimated to be within 100 m of the boundary between the Mount Lawson and Mount Moore Formations (see Fig. 2). Our sections through the Mount Lawson Formation are equivalent to the lower 500 m of section 3 in Hickey and others (1983).

Our MV section (Table 1 and Fig. 2) represents a 1,300-m section from the north coast of Matthew Peninsula. We estimate the base of MV to be about 100 m above the base of the Mount Moore Formation and a like amount below the top. As always, we chose the finest-grained material available. In order to achieve as dense sampling as possible, our section was pieced together from several shorter, well-exposed sections; lateral separation of these mini-sections did not exceed 100 m, and continuous exposure of prominent sandstone units allowed for unambiguous stratigraphic correlation among them. Our MV section is approximately equivalent to Section 1 of Hickey and others (1983).

The northern coast of Matthew Peninsula is structurally complex (see Fig. 2). The MV section was taken on the western limb of a plunging syncline, the nose of which is cut by at least one fault. In order to conduct a fold test, we took a short section, NEZ, at the nose of the syncline, where the strike of the beds is approximately perpendicular to the strike in the MV section. The NEZ section also spans the boundary between the Mount Moore and the Margaret Formations as recognized by previous field parties. The critical paleomagnetic section through the Margaret Formation was Section 2 of Hickey and others (1983). We examined this section carefully and could find no outcrops suitable for paleomagnetic sampling. The sediments are primarily unconsolidated sand or lignite and have suffered extensively from the effects of permafrost and frost heaving.

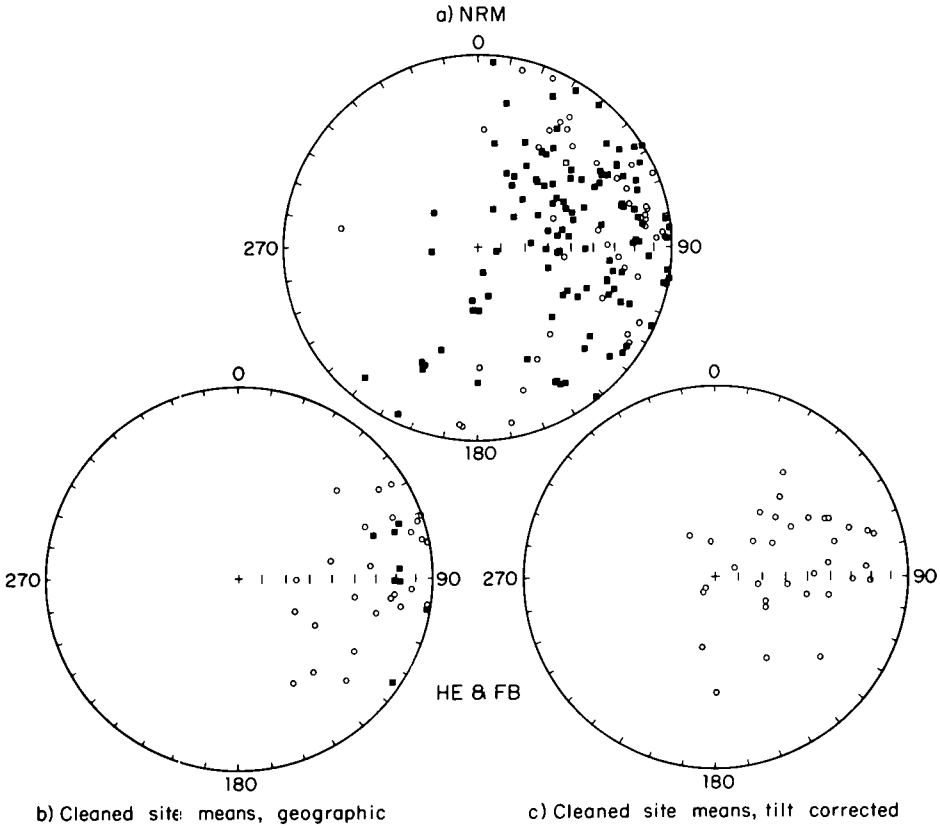



**Figure 4. Equal-area stereonet plots of specimen or site mean directions. Open symbols are projections onto the upper hemisphere (negative inclinations); solid symbols indicate the lower hemisphere (positive inclinations).**

## LABORATORY METHODS

Samples were prepared into paleomagnetic specimens by either sanding into cubes or setting in plaster and drilling out with a water-cooled drill press. Three to six specimens were obtained from every site. Two specimens from approximately every third site were selected to undergo detailed thermal and alternating field (A.F.) step-wise demagnetization. All demagnetization and measurement was carried out in the magnetically shielded room at Scripps Institution of Oceanography (ambient field of 200 nT or less than 1% of the Earth's magnetic field in San Diego). Schonstedt thermal and single-axis A.F. demagnetizers were used. All A.F. demagnetizations were carried out twice, in opposite orientations in order to minimize the effect of possible anhysteritic remanence. A vector mean of the two measurements was used for all illustrations and calculations.

All magnetic measurements were made on the Scripps CTF Inc., three-axis cryogenic magnetometer. Our procedure is as follows. The sample holder is measured, followed by a baseline measurement. The specimen is measured in four orientations with the orientation arrow on top, followed by four measurements with the sample inverted. The (very small) contributions of the sample holder and any components induced by the less than 20-nT field in the magnetometer are subtracted from the measurements. A final baseline measurement is taken to screen for drift or flux jumps and if drift has exceeded 30 $pAm^2$, the specimen is remeasured. Each reading consists of 25 separate measurements of each of the three axes. The standard deviation of the measurements is monitored for instrument noise (approximately 5 $pAm^2$), sample noise (caused by, for example, viscous decay), or operator-induced noise (for example, from attempting a measurement too soon after insertion for a particular slew rate and sample intensity). We calculate the circular standard deviation (CSD) of the 8 vector measurements, which rarely exceeds 1°–2° and is an indication of the reproducibility of measurement. These raw data are then corrected for sample and bedding orientation.

Typical results from representative specimens of both thermal and A.F. procedures are shown in Figure 3. In general, the specimens are remarkably well behaved, showing both normal (Figs. 3c and 3d) and reversed (all others) directions. In some cases, thermal demagnetization led to quite scattered results (Fig. 3h), owing to chemical changes at elevated temperatures. The characteristic direction is isolated by either 20 mT or 300 °C, and the trajectories generally (although not always) decay to the origin. The

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median destructive field is between 20 and 30 mT, and there is a maximum unblocking temperature between 575 and 600 °C, both consistent with a magnetite carrier. In this paper, we assume a magnetite remanence acquired as a depositional remanent magnetization (DRM) or postdepositional remanent magnetization (p-DRM) shortly after deposition.

Owing to the simplicity of remanence characteristics and the superiority of A.F. demagnetization, all specimens from the remaining sites were subjected to blanket demagnetization at 20 mT except specimens from the upper 300 m of the MV section. Each of these was subjected to stepwise A.F. demagnetization, owing to scatter in the blanket 20 mT data. The 300 °C temperature step was used for specimens that underwent step-wise thermal demagnetization.

After demagnetization, site means were calculated using Fisher (1953) statistics. Sites whose specimen directions were statistically random using the criteria of Watson (1956) were rejected for interpretation of polarity.

## RESULTS

The NRM data for all specimens and demagnetized, statistically significant site mean data are shown in Figures 4–6.

In Figure 4, we present data from the combined FB and HE sections. There is considerable scatter in the NRM data (Fig. 4a), with a preponderance of data in the Eastern Hemisphere. The significant site means of cleaned data are shown in geographic (Fig. 4b) and tilt-corrected coordinates (Fig. 4c). Means of the data before and after tilt correction are listed in Table 2.

As there is only one polarity in these data (reversed), no reversals test is possible to indicate the presence of an overprint. As a test for possible overprinting in the present (vertical) field, we calculated the principal component (Kirschvink, 1980) for all demagnetization data between 30 and 50 mT for each of the 10 specimens from the HE section subjected to stepwise treatment. After rejecting the directions from sites whose demagnetization data were nonlinear (maximum angle of deviation >10°), we calculated the mean of these linear (presumably univectorial) components. The mean direction (declination = 79.7, inclination = –56.6, N = 6, and $\alpha 95$ = 23) of the principal component data is identical to that obtained from blanket treatment (Table 2), suggesting that the data shown in Figure 4c accurately reflect the characteristic directions of the specimens.

The Late Cretaceous pole is well determined (see Van der Voo, 1981) and gives an expected paleofield direction of dec = 292° and inclination = 77° (95% confidence limits = 2°). The mean direction for the Mount Lawson Formation is some 20° too shallow, which could be explained as an inclination error acquired during deposition (Johnson and others, 1948), or flattening caused by compaction. It is, however, difficult to explain the mean direction in terms of overprinting or poor structural correction, although these are certainly possible explanations. In any case, the polarity of all sites is unambiguous.

In Figure 5, we show the vector data from the MV section. The NRM data clearly are affected by the present field (inclination +89°) and are streaked between a west-northwesterly and down direction and an east-southeasterly and up direction. The cleaned site means shown in Figure 5b have drastically reduced streaking. The grand means of geographic and tilt-corrected data are listed in Table 2. Data in geographic coordinates are assumed to be reversed if they are in the Eastern Hemisphere; tilt-corrected data are assumed to be reversed if they have negative inclinations. The normal and reversed data are antipodal (see Table 2), thus passing the reversals test.

Data from the NEZ section are depicted in Figure 6. The NRM data (Fig. 6a) are highly scattered; many of the sites had random within-site groupings. The cleaned data, selected by the criteria of nonrandom site mean statistics (shown in Figs. 6b and 6c), are well grouped and may be considered reversed.

The NEZ section stratigraphically overlies the MV section and was sampled primarily as a fold test. We therefore combined data from the two sections (N = 33). Intermediate directions (inclinations within the bounds of ±45°) tend to occur between fully normal and fully reversed zones (see MV data). Excluding these (transitional?) directions, we calculated dispersions before (K = 5.2) and after (K = 18.7) tilt correction. The dispersion is significantly decreased (99% confidence) after correction for tilt, strongly suggesting that the magnetizations were acquired prior to tilting. The timing of tilting is thought to have occurred prior to the early Oligocene (Miall, 1986). In the following interpretation, we will assume that the magnetization was acquired penecontemporaneously with deposition and accurately reflects the polarity of the field at or very close to the time of deposition.

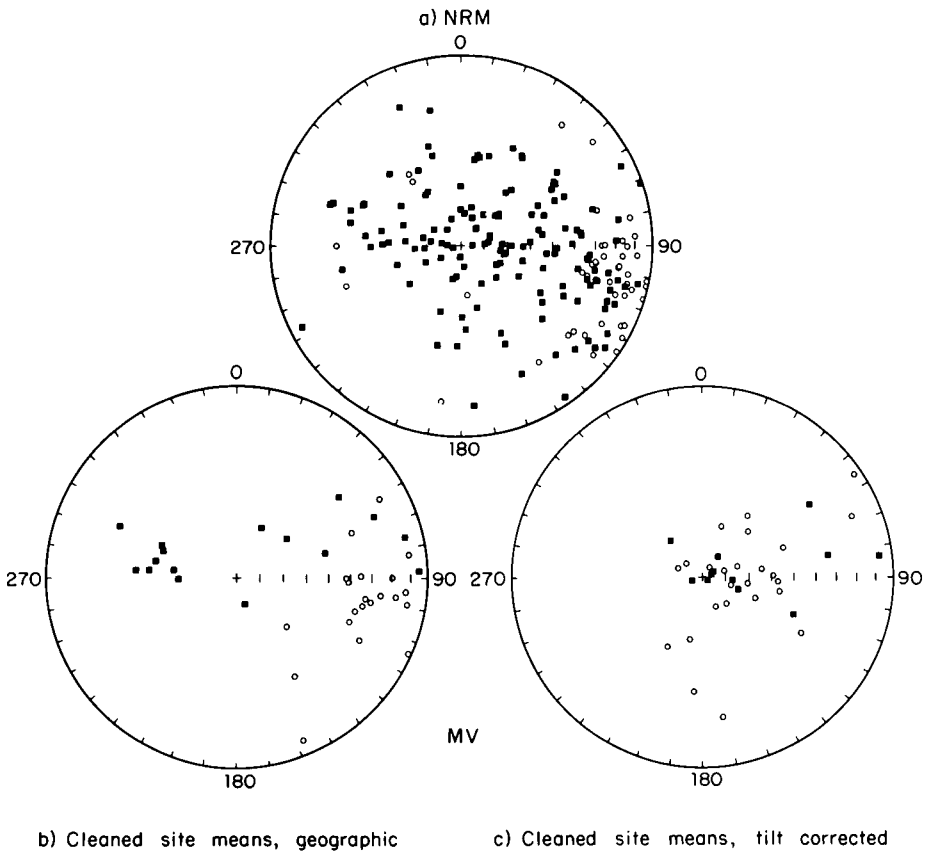



**Figure 5. Equal-area stereonet plots of specimen or site mean directions. Symbols same as Figure 4.**

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TABLE 2. STATISTICAL INFORMATION

| Section | Geographic Mean Decl. | Geographic Mean Incl. | N | K | α95 | Tilt corrected Mean Decl. | Tilt corrected Mean Incl. | N | K | α95 |
|---|---|---|---|---|---|---|---|---|---|---|
| FB + HE | 269.9 | 16.7 | 34 | 6.2 | 9.6 | 261 | 56 | 34 | 7 | 9.1 |
| MV (all) | 275.6 | 22.7 | 37 | 4 | 11.5 | 296.9 | 82.9 | 37 | 5.5 | 9.8 |
| MV (positive inclination) | | | | | | 77.3 | 69.0 | 12 | 6.9 | 15.4 |
| MV (negative inclination) | | | | | | 278.8 | -70.7 | 25 | 6.9 | 10.6 |
| MV (without intermediate) | 275.4 | 34.1 | 28 | 12.8 | 7.4 | 272.2 | 81.2 | 28 | 17.1 | 6.4 |
| NEZ (all) | 284.3 | 66.9 | 7 | 6 | 21.7 | 230.2 | 73.0 | 7 | 5.7 | 22.1 |
| NEZ (without intermediate) | 240.2 | 77.7 | 5 | 37.3 | 10.2 | 353.8 | 74.2 | 5 | 62.6 | 7.9 |
| MV + NEZ (without intermediate) | 282.5 | 47.8 | 33 | 5.2 | 10.7 | 265.8 | 80.9 | 33 | 18.7 | 5.6 |

## MAGNETOSTRATIGRAPHY

The magnetostratigraphy of our sections is shown in Figures 7 and 8. In Figure 7, we show the combined results of the FB and HE sections. Stratigraphic correlation is based on the lateral tracing of a prominent, thin sandstone which is continuously exposed between the two sections. We plot the tilt-corrected inclination data for all specimens as small circles. The solid lines connect mean inclinations of sites with nonrandom directions (squares). We interpret this section to be entirely reversely magnetized.

The magnetostratigraphic results from the Matthew Peninsula are shown in Figure 8. As before, we plot the inclinations of the individual specimens as well as the nonrandom-site mean inclinations. The MV section is dominantly reversed but has a prominent, ~300-m-thick normal zone in the middle; a short normal zone immediately above the thick one, represented by only one site with a low inclination; and a normal zone (2 sites) at the top associated with a zone containing highly scattered data. The thickness of the uppermost normal is poorly constrained. We consider the upper two normal zones to be tentative, whence the question marks on the polarity log.

The correlation of the NEZ section to the MV section is somewhat problematical, as there is at least one fault separating the two areas. The Mount Moore/Margaret formational boundary, however, has been mapped in both areas, and, if this is a reliable stratigraphic marker, then the two sections are correlated as in Figure 8.

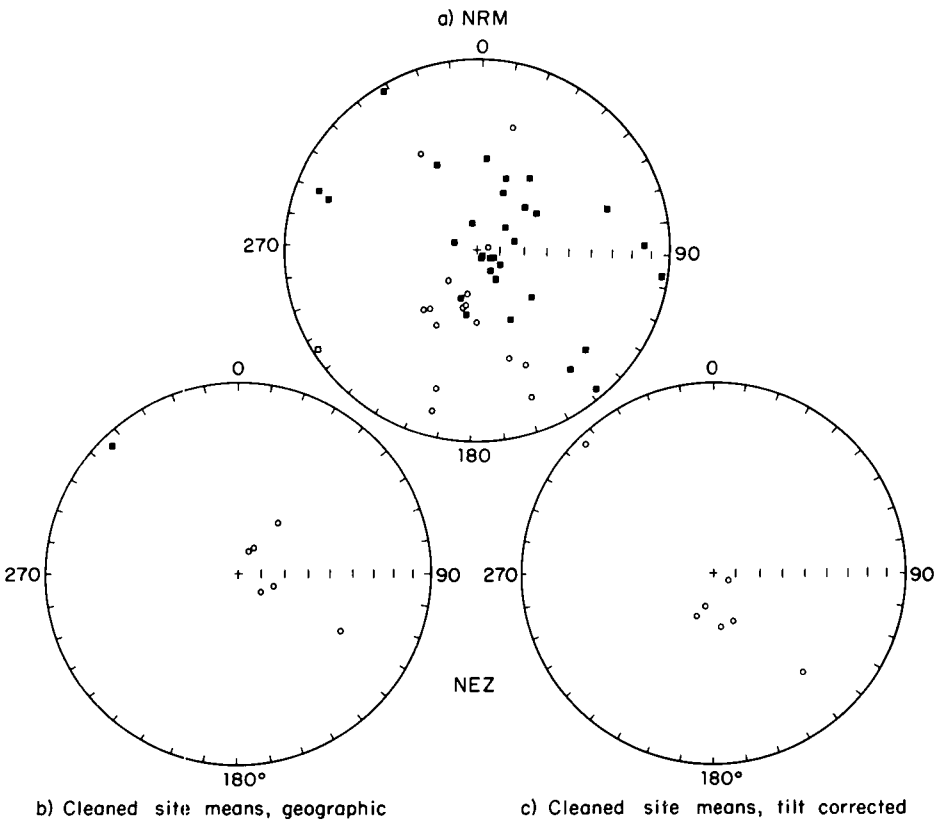



**Figure 6. Equal-area stereonet plots of specimen or site mean directions. Symbols same as Figure 4.**

## DISCUSSION

### Comparison with Prior Results

In Figure 9, we plot the magnetostratigraphy published by Hickey and others (1983). According to their correlation, the lower and upper vertebrate localities (late Wasatchian and perhaps early Bridgerian) are associated with anomalies 25 and 23, respectively (that is, subchrons C25N and C23N). The magnetostratigraphic data presented here are shown to the right of the prior results. We used the formational boundaries mapped by previous expeditions as our frame of reference in the field, and these can therefore be used for direct comparison of the two data sets. Stratigraphic thickness data, however, were measured by the two groups separately and may differ, depending on assumed bedding attitude and precise location. For example, we measured 1,500 m of the Mount Moore Formation on the northern coast of the Matthew Peninsula, whereas the stratigraphic thickness of this formation is shown as about 1,200 m from measurements on the south coast of the Matthew Peninsula in Hickey and others (1983). Such differences are within the error of the measuring technique.

There are notable differences between the two composite magnetostratigraphies. Part of the cause for discrepancy is that we sampled more cautiously, avoiding poorly lithified and sandy units; we therefore have very few data for the Margaret Formation and a 200-m gap between the Mount Lawson and Mount Moore Formations. Moreover, where there is overlap, there is little correspondence between the two data sets. For example, our section has a total of three normal polarity zones (two of which are considered tentative). The data used by Hickey and others (1983) through the same interval suggest a minimum of nine normal zones, which they

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correlated to anomalies 33 through 26. One possible explanation for the difference is that all of our polarity data are based on site means of from three to six individual specimens, rejecting sites with random within-site groupings. The prior data set was apparently based on single samples from each stratigraphic level. Thus, many polarity zones shown in the previous composite correspond to levels with scattered data in our sections and hence have not been interpreted by us in terms of polarity (see Figs. 7 and 8). It is also possible that magnetic shielding of the laboratory environment at Scripps helped reduce the scatter of the data presented in this paper.

Whatever the cause of the discrepancy, the arguments of Kent and others (1984) against the overall quality of the former data set and the arguments presented here for the reliability of the new data set suggest that the polarity log presented here should supersede that used by Hickey and others (1983).

Using the correlation of Hickey and others (1983), there is an implied duration of ~30 m.y. for the section resampled by us, suggesting an average sediment accumulation rate of around 67 m/m.y. Our magnetostratigraphy, regardless of correlation, virtually precludes such a long period of continuous sedimentation. One is forced to conclude that either there are long, undetected, and serendipitously located hiatuses which cut out many normal zones, or the time span of the section is very much shorter than 30 m.y. The longest post-Permian reversed interval is about 5 m.y., which severely constrains the duration possible for the sections considered here. The deltaic facies as described by Miall (1981) argue for overall higher sediment accumulation rates than implied by Hickey and others (1983), an argument also supporting a shorter duration for our sections.

## Implications for Faunal Heterochroneity

We plot the paleomagnetic and mammalian faunal data from the Clark's Fork Basin of Wyoming (Butler and Lindsay, 1985) and from Big Bend National Park (Rapp and others, 1983) in Figure 9. The Wasatchian localities in the Clark's Fork Basin are in a reversed zone identified as Chron C24R. Those from Big Bend come from a formation directly overlying the sampled section and are therefore younger than C24R. Localities in the Eureka Sound Formation that have fossils which correlate best to the Wasatchian levels defined in the western United States are above our sampled section. The Eureka composite magnetostratigraphy presented in this paper, however, can reasonably be correlated with that from the Clark's Fork and Big Bend sections. There is, therefore, no paleomagnetic evidence for a significant time difference between the Wasatchian fauna from Ellesmere and those of the classic western United States localities as postulated by Hickey and others (1983).

## Implications for Floral Heterochroneity

The most significant temporal discrepancy between the Canadian Arctic and the southwestern United States postulated by Hickey and others (1983) was in the pollen record (data quoted from Choi, 1983). The Mount Lawson Formation (Member II) is characterized by pollen zone B, which is associated with Paleocene strata in the southwestern United States. This zonation is based on two localities (67 and 71 of Choi, 1983) in section 3 of Hickey and others (1983), resampled by us as the FB and HE sections. The Mount Moore Formation (Member III) was identified as Zone C, based on locality 66 of Choi (1983) taken from Section 1 of Hickey and

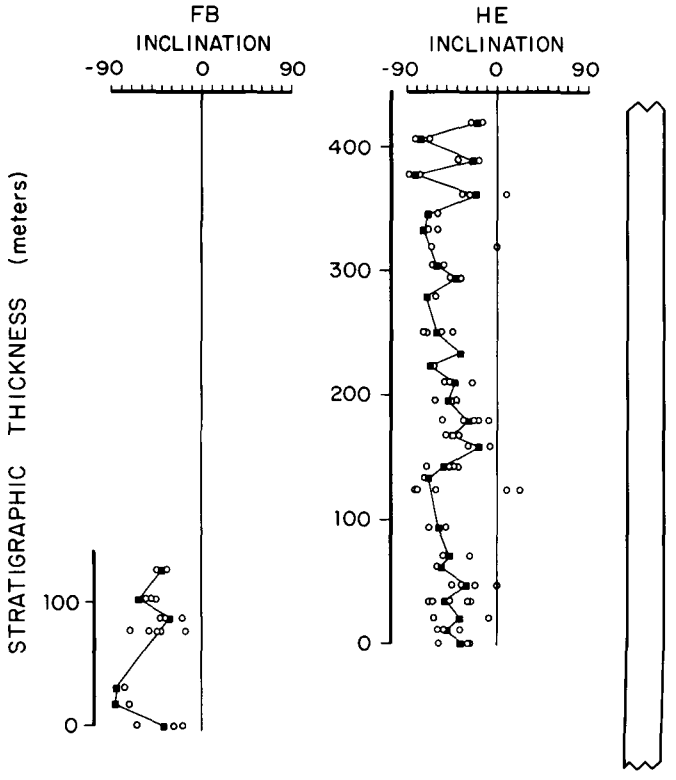



**Figure 7. Inclination data of the FB and HE sections (Mount Lawson Fm.) in stratigraphic order. Open symbols are the individual specimen inclinations, and solid symbols connected by lines are inclinations of statistically significant site means. The polarity interpretation is shown to the right and is entirely reversed.**

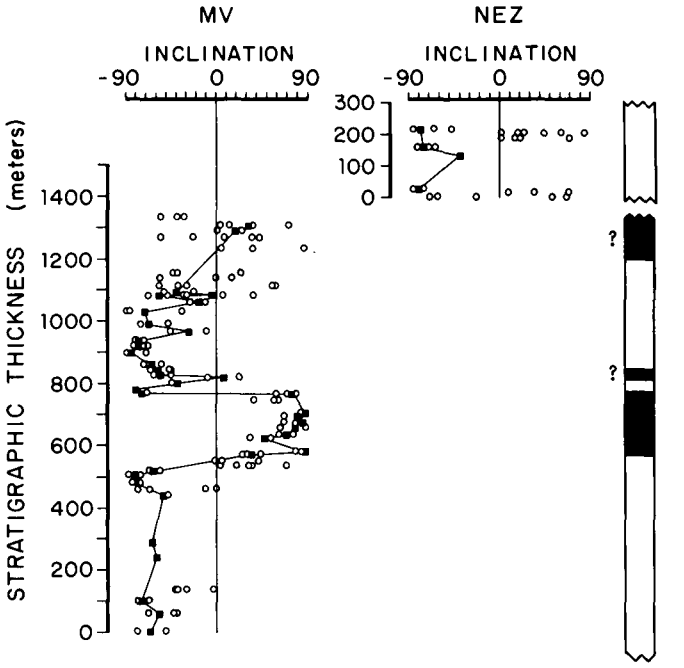



**Figure 8. Inclination data from the MV (Mount Moore Fm.) and the NEZ (Mount Moore/Margaret Fms.) sections. Symbols same as Figure 7. Polarity log to right shows normal (black) and reversed (white) interpretations based on the sign of the mean inclination data.**

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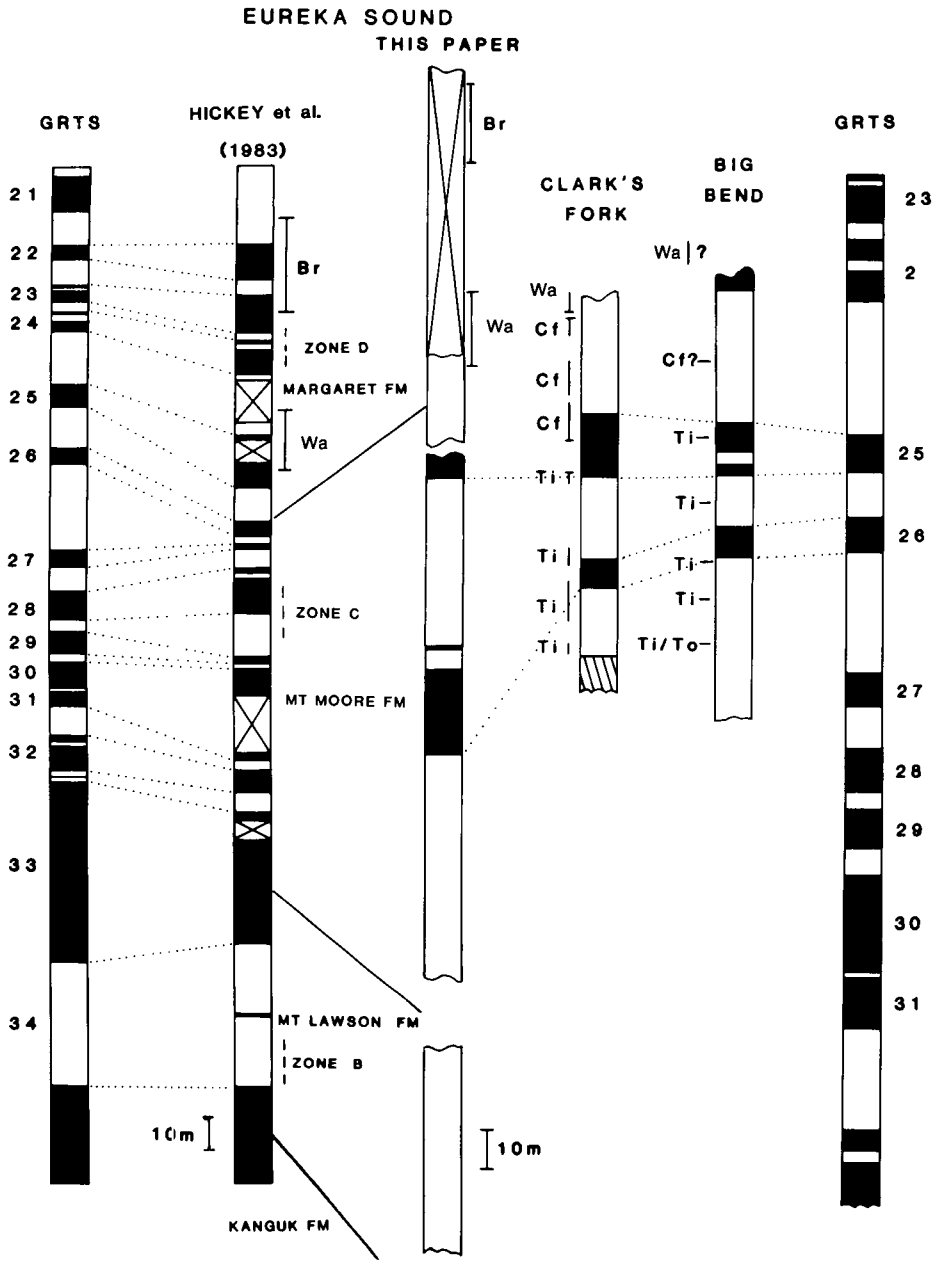



**Figure 9. Comparison of the data presented here with that by Vinson (1981; published by Hickey and others, 1983). The previous correlation of the Eureka Sound sediments to the Geomagnetic Reversal Time Scale (GRTS) is shown to the left. The magnetostratigraphic data from Clark's Fork (Butler and Lindsay, 1985) and Big Bend (Rapp and others, 1983) as correlated to the GRTS are shown to the right. Dotted lines indicate temporal correlations; solid lines indicate stratigraphic correlations of formational boundaries. Fossil horizons identified in terms of Land Mammal Age affinities are shown in stratigraphic position. Abbreviations: To, Torrejonian; Ti, Tiffanian; Cf, Clark-forkian; Wa, Wasatchian; Br, Bridgerian.**

others (1983). Zone C is thought to be late Paleocene in the Southwest. The two localities of Choi (1983, localities 50 and 51) in the Margaret Formation yield Zone D pollen, which compares with Eocene pollen in the Southwest. The approximate locations of the pollen localities and their zonal designations are shown in Figure 9 (data from Hickey and others, 1983, and Choi, 1983).

The contention that the flora of the lower Eureka Sound Formation (particularly Members II and III) preceded that in the Southwest by as much as 18 m.y. was based on the association of Zones B and C, dated elsewhere as Paleocene, with Campanian strata. The postulated Campanian age for the Mount Lawson Formation (Member II) was based on the recognition of Chron C33N ("long magnetic normal interval 33") in Section 3 of Hickey and others (1983). This correlation was supported by late Santonian to Campanian age molluscs found in Member I facies on the Fosheim Peninsula some 150 km to the north (see Fig. 1). We found no evidence for a long magnetic normal zone in our resampling of the sequence. Furthermore, correlations of deltaic facies across such large distances are unlikely to have temporal significance. Therefore, the argument that Zone B is actually Campanian rather than Paleocene does not have convincing support. To the contrary, assigning a Campanian age to the Paleocene-like flora appears inconsistent with the magnetostratigraphic and lithostratigraphic data presented here. If true, there must either be an 18- to 20-m.y. hiatus in sedimentation not evident from field observations, presumably at the Mount Lawson/Mount Moore formational boundary, or the entire sequence is Campanian, suggesting an even more substantial temporal discrepancy than heretofore postulated. On balance, the simplest interpretation of the combined data presented in Figure 9 is one of little or no temporal discrepancy between the Arctic and the southwestern United States faunas and floras.

## CONCLUSIONS

1. Sediments of the Eureka Sound Formation near Bay Fiord are suitable for paleomagnetic analysis. Remanence is most likely carried by magnetite, and the characteristic remanence is isolated in most cases by alternating field demagnetization to 20 mT. A positive fold test suggests acquisition of remanence close to the time of deposition. Sediments of the Mount Lawson Formation (Member II) have, on average, significantly shallower directions than those of the overlying Mount Moore Formation (Member III). This may be the result of errors, occurring in one facies and not the other, caused by compaction or depositional processes.

2. We were unable to duplicate the magnetostratigraphic results of Vinson (1981) and published by Hickey and others (1983). These results were the principal basis for the hypothesis that Arctic flora and fauna predated similar populations to the south by as much as 18 m.y. Such a temporal discrepancy is not supported by the data presented here. To the contrary, comparison of the Ellesmere magneto-biostratigraphic data to those from the Clark's Fork Basin (Butler and Lindsay, 1985) and the Big

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Bend area (Rapp and others, 1983) show no remarkable differences. On the basis of the present data set, then, we cannot exclude the hypothesis that the Ellesmere mammalian and plant populations lived contemporaneously with similar faunas and floras to the south.

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